C8.4 Barrier requirements

This section describes the formal requirements for barrier transactions.

C8.4.1 Master requirements

For a master component issuing a barrier transaction, the following rules apply.

- Both transactions in a barrier pair must have the same **AxID**, **AxBAR**, **AxDOMAIN**, and **AxPROT** values
- If the **ARID** and **AWID** signals have different widths, the narrower version must be zero-extended to match the wider version
- Barrier pairs must be issued in the same sequence on the read address and write address channels
- A master interface is not required to issue barrier transactions on the read address and write address channels in the same cycle
- A master interface is permitted to issue multiple outstanding barriers, meaning that additional barrier transactions can be issued before responses to earlier barrier transactions are received. However:
  — an ACE-Lite master interface can issue outstanding barrier transactions without restriction
  — an ACE master interface must not issue more than 256 outstanding barrier transactions.

  Note

  Read and write response handshakes are separate events that can occur in any order. Therefore, a barrier is defined as an outstanding barrier from the cycle when the first of the read or write barrier becomes valid until the cycle when both the read and write response handshakes have occurred.

- Barrier transactions are required to use different ID values than those used for non-barrier transactions. It is permissible for barrier transactions and non-barrier transactions to use the same AXI ID value, provided one transaction has completed before the other is issued.

  Note

  Using different ID values ensures that any component tracking barrier responses does not have to track all responses to differentiate between a barrier response and a normal transaction response.

On each address channel, any transaction issued before the barrier on that channel is defined to be before the barrier, even if it is issued after the corresponding barrier on the other address channel.

Figure C8-3 shows pre-barrier and post-barrier transactions, with respect to barriers being issued on the address channels and responses being received.

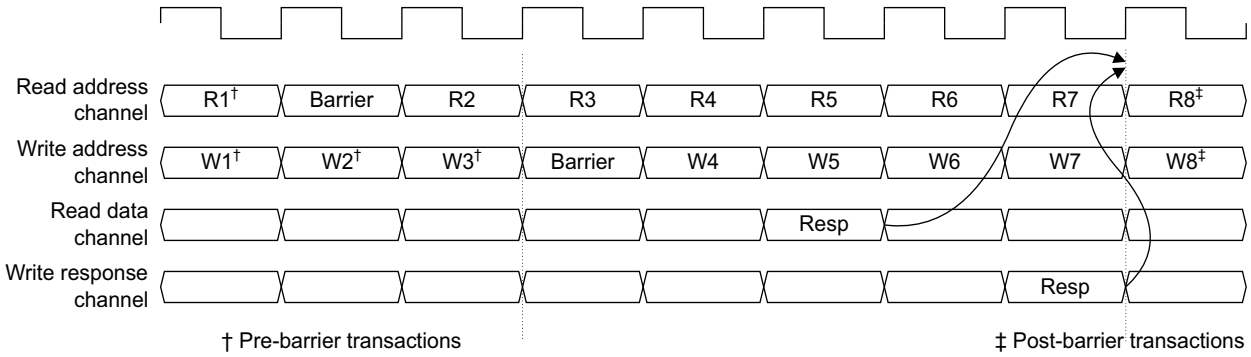


Figure C8-3 Barrier transaction timing

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In Figure C8-3 on page C8-254:

- transactions R1, W1, W2, and W3 are before the barrier
- transactions R8 and W8 are after the barrier
- transactions R2, R3, R4, R5, R6, R7, W4, W5, W6, and W7 have no relationship to the barrier.

The following rules apply to master components issuing barrier transactions, and relate to non-barrier transactions:

- A master must not issue any transaction, either read or write, that must be after the barrier until the master has received a response for the barrier on both the read data and write response channels.
- A master is permitted to issue transactions between issuing a barrier transaction on the address channel and receiving the read and write barrier responses. Such transactions have no ordering guarantees with respect to the barrier. On the address channel, these transactions are permitted to remain after the barrier transaction or they are permitted to overtake the barrier transaction.
- A master interface that has issued a read barrier on the read address channel must issue the corresponding write barrier on the write address channel, in a timely manner, if all other transactions on the write address channel are progressed. The master interface must not require either handshaking or a response to the read barrier or any read transaction after the read barrier before issuing the corresponding write barrier.
- A master interface that has issued a write barrier on the write address channel, must issue the corresponding read barrier on the read address channel, in a timely manner, if all other transactions on the read address channel are progressed. The master interface must not require either handshaking or a response to the write barrier or any write transaction after the write barrier before issuing the corresponding read barrier.
- A barrier must not be issued on the read address channel if subsequent read transactions are required for either:
  - issuing the corresponding barrier on the write address channel
  - issuing any write transactions that must be before the barrier.

  For example, a read barrier must not be issued if, after issuing the read barrier, it is necessary to perform translation table walks to issue write transactions that must be before the corresponding write barrier.
- For an ACE-Lite interface, a barrier must not be issued on the write address channel if subsequent write transactions are required for either:
  - issuing the corresponding barrier on the read address channel
  - issuing any read transactions that must be before the barrier.
- For an ACE interface, a barrier must not be issued on the write address channel if subsequent write transactions that must be ordered with respect to the barrier, must be issued for either:
  - issuing the corresponding barrier on the read address channel
  - issuing any read transactions that must be before the barrier.
- An ACE interface is permitted to issue a write barrier, followed by any of the following transactions that are required for snoop transactions to that master to complete:
  - WriteBack
  - WriteClean
  - Evict

The following rules apply to ACE master components and the interaction of barriers and local cache accesses:

- A master must not perform a store that must be ordered with respect to the barrier, to a shareable location in its local cache until after the barrier response is received on both read data and write response channels. This rule applies even if there is no requirement for a transaction to be issued because the cache line is in a Unique state.
- A master must not perform a load that must be ordered with respect to the barrier, from a shareable location in its local cache until after the barrier response is received on both read data and write response channels. This applies even if there is no requirement for a transaction to be issued because the cache line is in a Valid state.

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- A master must be capable of issuing write transactions to complete snoop transactions, even if the read address channel is stalled.
- Issuing a barrier transaction must not prevent any of the following transactions, that are required for snoop transactions, being issued and completed:
  - WriteBack
  - WriteClean
  - Evict

C8.4.2 Slave requirements

The following rules apply to a slave component that is handling barrier transactions:

- On receipt of a barrier transaction, an ACE-Lite slave interface is permitted to either:
  - stall the read address channel until it receives the corresponding barrier transaction on the write address channel
  - stall the write address channel until it receives the corresponding barrier transaction on the read address channel.
- An ACE slave interface must be able to accept 256 barrier transactions on the write address channel without blocking the progress of subsequent transactions. It is required that the write address channel is available and that write transactions can progress.
- On receipt of a read barrier, an ACE slave interface is permitted, but not required, to stall the read address channel.

C8.4.3 Interconnect requirements

The following rules apply to an interconnect processing barrier transactions:

- The interconnect topology must not permit transactions with overlapping destination addresses to have a common start point and end point but have different paths through the interconnect.
- When merging two streams of transactions, an interconnect must ensure that barrier pairs are issued on the read and write channels in the same sequence. Barriers pairs must not be interleaved.
- Any interconnect component that has multiple ACE slave ports must be capable of meeting the ACE slave interface requirements in *Slave requirements*, for all ports simultaneously.
- An interconnect must not permit a barrier transaction to overtake any transaction that respects barriers.
- A barrier must apply to any transaction that the component issuing the barrier observed before issuing the barrier.
- An interconnect that has not responded to a barrier can permit any non-barrier transaction to overtake that barrier.

Note

This specification recommends that ACE interconnect components stall a read barrier until:

- the corresponding write barrier is received
- all transactions before the read barrier and write barrier have been snooped as required, and all write transactions that must be before the barrier have been issued.

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C8.4.4 Barriers and Device transaction ordering

Barrier transactions ensure ordering between Device transactions to a single peripheral device, regardless of the addresses within the peripheral being accessed. This means that any hazard-checking that an interconnect performs when responding to a barrier must be extended to the entire address space of the peripheral, and must not consider only overlapping transactions. If an interconnect is unable to determine the address range of a particular peripheral, it must ensure ordering between all accesses that could be addressing that peripheral.

C8.4.5 Multi-copy atomicity requirements for shareable locations

For Inner Shareable and Outer Shareable locations, multi-copy atomicity is required. This means that a snoop response to a write is issued only when all observers in the required shareability domain have observed the write. Also, on the cycle that a snoop response is received, the associated master must have already observed the write. The point of observation is defined as the handshake on the snoop response channel. The snoop data channel is not used in defining the point of observation. This means there is no requirement for barriers on the snoop address channel.

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Chapter C9
Exclusive Accesses

This chapter describes ACE Exclusive accesses. It contains the following sections:

- *About Exclusive accesses* on page C9-260
- *Role of the master* on page C9-261
- *Role of the interconnect* on page C9-263
- *Multiple Exclusive Threads* on page C9-266
- *Exclusive Accesses from AXI components* on page C9-267
- *Transaction requirements* on page C9-268.

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